Memo for “Physical Appearance and Elections: An Inequality Perspective”

We thank the Editor for giving us the opportunity to revise our work and the two reviewers for their comments, suggestions, and excellent constructive criticism. **We have seriously considered every comment and suggestion and sought to implement all of them in our revision**. We hope that the revised paper reads better and shows substantive and methodological improvements.

We have structured the memo in two parts. In the first part, we respond to Reviewer #1 (hereafter, R1), and in the second part, we respond to Reviewer #2 (hereafter, R2). To enhance the efficiency of this memo, we have restated the original question (in boldface for greater readability) before answering. Immediately after that, we show the relevant paragraph that will go into the paper (in blue for readability). We have also included references to the edited version to ensure that the reader can cross-check our revisions.

Reviewer #1

We want to thank you for your comments and suggestions. We believe your constructive criticism definitely improved our paper. Below you will find that we have done all we could to implement **all** your suggestions.

1. **What makes somebody look like having a working class, middle class or upper class occupation? In other words, is there any criterion that makes somebody either working, class, middle class or upper class?**

   Thank you very much for this question. We think this is a great opportunity for us to clarify even further.

   We have taken three measures to answer your question. First, we have incorporated an edited paragraph explaining our measurement strategy. As we explain in that section (see relevant paragraph below), we did not ask respondents to evaluate the degree to which candidates’ physical appearance was congruent with their social class. Since, in the Finnish context, this constitutes an inappropriate question (see detailed explanation below), social desirability biases could have contaminated the answers we would have received. Second, we clarify that we asked respondents instead whether the candidates’ physical appearances were congruent with their occupations. As the paper notes, candidates state their occupations in the poster campaigns. Importantly, and Third, and importantly, we took a data-driven approach: Raters did not receive instructions or definitions of how a “janitor” or a “lawyer” should look. Thus, for us, this is rather an empirical question. As we note in the edited paragraph we added after reading your comment, we now clarify that our data-driven approach consisted of evaluating “**the extent to which a member of a category (in this case, a person representing a certain occupation) fits a more generally shared idea or image of the meaning of the category name (cf. Rosch**

& Mervis, 1975)" (p. 8 of the edited paper). This approach then (empirically) exploits these shared ideas without defining, **neither *a priori* nor directly**, what "upper," "middle," or "working" class means.

Thus, we have added the next paragraph (p. 8 of the edited paper) addressing this issue:

"Ideally, we would have directly asked the 7,920 raters to evaluate whether candidates looked as if they came from the working, middle, or upper class. However, that would have introduced social-desirability biases into our responses. This widely studied effect appears in other contexts where subjects avoid answering truthfully due to social pressure, especially when the questions ask about drug use (Druckman et al., 2015), sexual preferences (LaBrie & Earleywine, 2000), attitudes toward race (Kuklinski et al., 1997; Redlawsk et al., 2010) or vote buying (Blair & Imai, 2012; González-Ocantos et al., 2012; Bahamonde, 2022). We believe that directly asking raters to make inferences about the candidates' social classes based on physical appearances constituted an inappropriate question in the Finnish context. Furthermore, the term "social class" is rarely used among the general population because of the "equal Finland" narrative. Thus, to elicit truthful answers, our empirical strategy considered instead asking this question indirectly, that is, asking raters to evaluate the extent to which a member of a category (in this case, a person representing a certain occupation) fits a more generally shared idea or image of the meaning of the category name (cf. Rosch & Mervis, 1975), only then matching those occupations within the three social classes."

In addition, and very importantly, we have added a theoretical and an empirical discussion about the relationship between social class and physical appearance. These paragraphs were added to the edited version of the paper (on p. 4):

"An alternative approach that links physical appearance and inequality is the Bourdieusian approach. According to this framework, social classes are based on different types of capital: economic capital (e.g., wealth and income), cultural capital (e.g., educational qualifications, attitudes toward art, culture, clothing, and physical appearance), and social capital (e.g., social relations), the composition and volume of which varies from one social class to another (Bourdieu, 1984, 1986). Thus, for example, the upper class might comprise individuals with relatively higher levels of cultural capital but not excessive incomes (e.g., professors), as well as people with higher incomes but not necessarily much formal education (e.g., successful entrepreneurs; Flemmen et al., 2017). Even more importantly, Bourdieu emphasized the embodied nature of social class: clothing and physical appearance can provide social markers that sort different social classes, according to how individuals dress and look (Bourdieu, 1984).

All in all, the Bourdieusian approach consistently makes the case that social class is embodied. Thus, people not only express their own social class through their appearance but also make inferences about other people's class status based on the way they look (Kuipers, 2015). Thus, styles of grooming (i.e., make-up, hair, and beard styles), clothing (Skeggs, 2001; Nelissen & Meijers, 2011; Kraus et al. 2013), and body size and shape (Keas & Beer, 1992; cf. Vandebroeck, 2016), become cues

of social class. Importantly, such cues make a difference, for instance, in "filtering in" certain job candidates in some labor markets (De Keere, 2022; Friedman & Laurison, 2020; Mears, 2011). Furthermore, recent research has shown that young children can already classify other people into occupational status categories using attributes of physical appearance as cues (Vandebroeck, 2021). We believe a similar process of occupational status recognition and "filtering in" likely takes place in low-information elections."

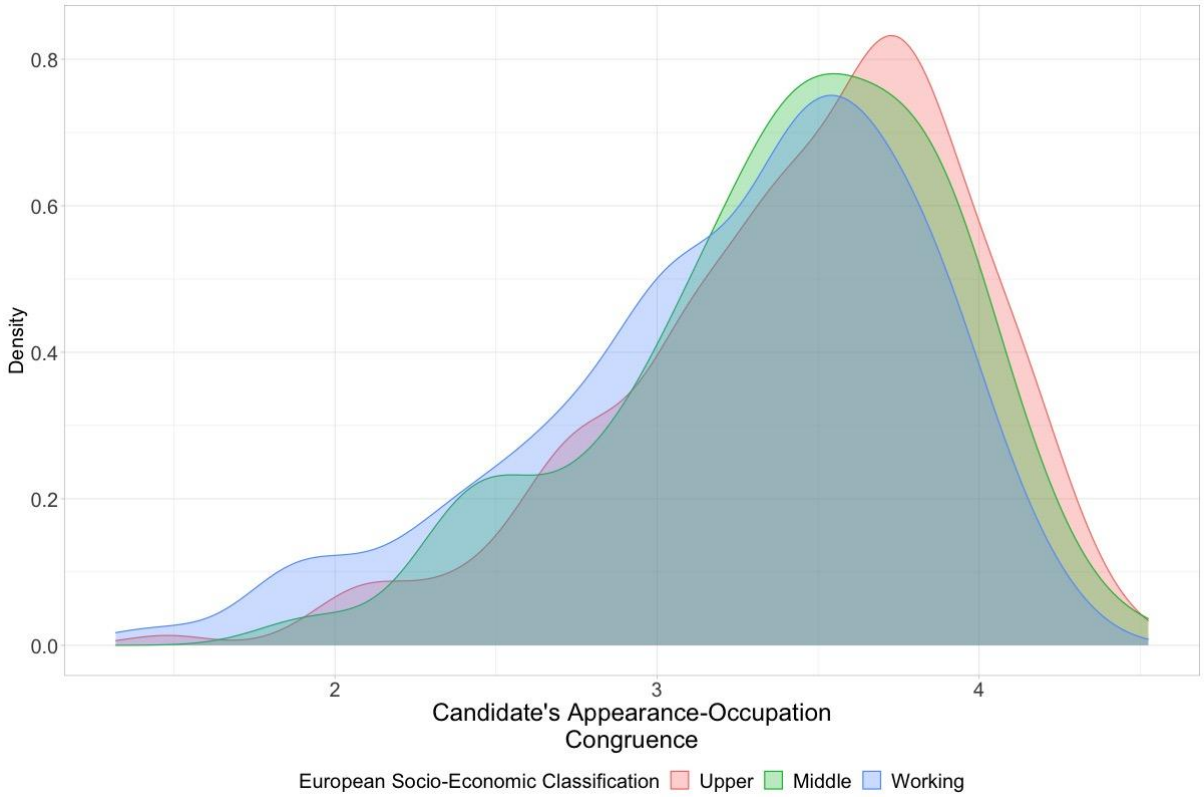
We hope that these added paragraphs clarify the issue. We want to thank you once again for this comment. If there is anything else we should explain and/or incorporate in our paper, we would gladly do so.

2. **I wonder how we can detect the class status of candidates. On official candidate photos, they should all look alike. Men are probably in a suit and women in a dress. How can we detect the occupational status from these features?**

   Thank you so much for this comment; we think it is highly relevant. You are absolutely right: One would think that all candidates "should look alike." However, that is not the case.

   To address your point in full, we have taken several measures. First, we have added some discussion where we explain (based on past research—see the relevant paragraph below) that due to the electoral institutions that rule Finnish Municipal Elections, parties are usually forced to recruit as many candidates as possible, effectively getting "normal"/"ordinary" people on board and not just "career-oriented" politicians. This fact actually shows in the poster campaigns, as candidates (both men and women) dress in a very casual way: it is very easy to see suits, hoodies, knitwear, etc. Second, we now highlight (also based on past research; see relevant paragraph below) a feature of Finnish society that is heavily based on functionality and frugality, which, for the same reasons, does show in the candidates' pictures.

   Thus, in sum, not all candidates look alike. On the contrary, Finnish citizens actually can distinguish working-, middle-, and upper-class "looks" because they *do* look different from each other. As **Figure 1** shows on p. 10 (which was in our original submission), particularly in **Panel B** (see panel detail below), Finns effectively and consistently can tell the difference between working-, middle- and upper-class-looking candidates. (The three densities show high levels of convergence toward the mean, with little departure from normality.)



In particular, Panel B of Figure 1 (a figure already included in the original submission of our paper) shows that the variances are quite consistent among working-, middle-, and upper-class-looking candidates. Also, the figure shows that the mean of these congruence scores is approximately 3.5 (out of a max of 5—see Table 1 on p. 9 for more details about our 1–5 scale of congruence). All in all, (1) there are differences among candidates, and (2) Finnish citizens can identify them.

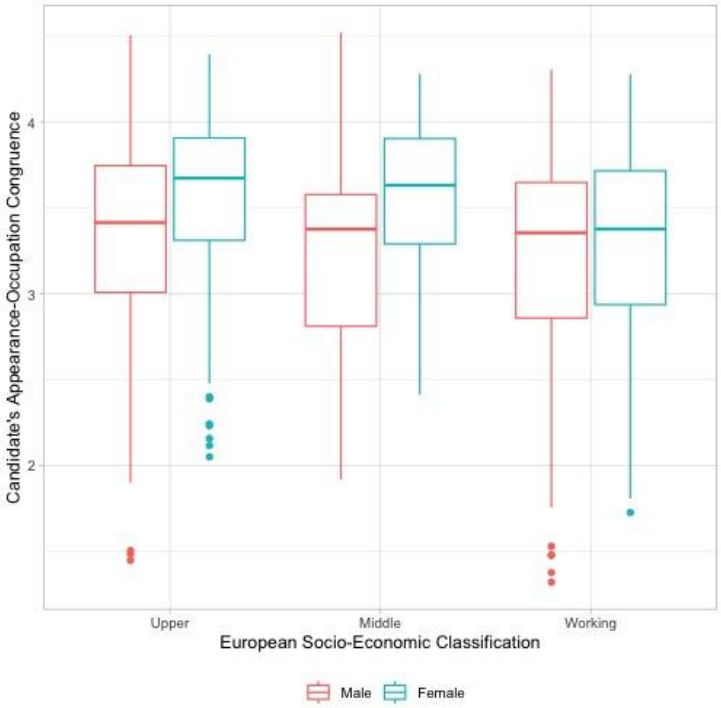
Added paragraph on p. 7 where we explain why there are "normal" and diverse "looks" among candidates:

"Finnish municipal elections use the D'Hondt electoral system, in which citizens vote for candidates running on party lists. The votes candidates receive determine their rank. Seats are then filled in proportion to the total votes received by each party list. Consequently, parties have strong incentives to present large lists of candidates, even if that means bringing in "list fillers." Importantly, recruiting such a large number of candidates to fill in the lists is a difficult task for parties, which must actively convince local residents to stand as candidates (Kestilä-Kekkonen et al., 2018). Thus, recruited individuals are not necessarily career-oriented politicians. As a result, ordinary people do run for municipal elections.[1] This electoral feature is critical for our identification strategy as it ensures that "normal" individuals do campaign, as opposed to, e.g., only the wealthy or the good-looking. Notably, this feature shows in campaign posters, where casual knitwear and even hoodies are as common as suits

[1] However, the nomination process might differ significantly between larger and smaller municipalities (Kestilä-Kekkonen et al., 2018). For example, for the 2017 Finnish Municipal Elections, Helsinki had 7,500 inhabitants per counselor, while in smaller municipalities the ratio was around 50 inhabitants per counselor (Borg & Pikkala, 2017).

and other more formal clothes, a fact that goes hand-in-hand with the Finnish emphasis on functionality and frugality (Sarpila, Kukkonen et al., 2021).[2]"

An important additional measure to address your question was to develop a new figure (Figure 2 in the edited version of the paper, on p. 11):



The figure shows that (1) candidates look like the occupation (social class) they have, and (2) that Finnish citizens can correctly distinguish these differences.

We also added the next paragraph, which expands on these two points.

Added paragraph that addresses the issue of "[how to] detect the class status of candidates [because all candidates] should all look alike."

"For internal consistency, we would also like to see Finnish voters perceiving working-class candidates as working-class individuals, middle-class candidates as middle-class individuals, and so on. Similarly, we would like to see that Finnish candidates who look as though they have working-class occupations actually are working-class individuals, candidates who look as though they have middle-class occupations actually are middle-class individuals, and so on.[3] Other results would mean that either Finnish voters have a hard time assessing the occupation (class) of the candidate based on the candidate's appearance, or candidates systematically look different from what they are. Either way, one or both assumptions failing to hold would seriously compromise our empirical claims.

[2] We thank one of the reviewers for encouraging this discussion.

[3] We thank one of the reviewers for encouraging this discussion.

To verify these assumptions, we show in Figure 2 that Finnish candidates scored an average of 3.3 (see Table 1; a score of 5 is the maximum congruence between physical appearance and occupation). This implies that Finnish voters find that 66% of the time, candidates do look like they belong to the class to which they actually belong. The figure also shows that candidates who look like they have working-class occupations are actually working-class individuals, and so on. Importantly, these levels are consistent for both male and female candidates."

We want to thank you once again for this comment. In fact, we made sure to thank you in a footnote as well (footnote #9 in the paper) for encouraging this discussion. If there is anything else we should explain and/or incorporate in our paper, we would gladly do so.

4. **What is the inter-coder inter-reliability in this study? In other words, how many coders converge in their occupation judgment? For this analysis to be trustworthy there needs to be strong convergence in voters' evaluation**.

This is another great question, and we are sorry if the original manuscript was unclear regarding our procedures.

We did not use "coders" in the typical sense. Coders are usually convenient samples (research assistants, graduate students, etc.) paid to code. As we call them in our paper, we effectively recruited 7,920 "raters." They are a random sample of the Finnish-speaking Finnish population aged 18–64. To address R1's question more directly, as **Table 1** states on p. 9 (this table was in the original submission and is pasted below for convenience), raters were divided into five samples to avoid **fatigue effects**. Since it was not advisable to ask each of our raters to evaluate all 1,500 candidate images, raters were divided into five subsamples. Each sample evaluated one of the five attributes (see details below in Table 1).

Statistically speaking, having raters randomly drawn from a population should take care of all observable *and* unobservable rater characteristics (e.g., gender, political attitudes, preconceptions of what, say, "working-class looking" means). In other words, our representative sample of Finns (i.e., the raters) and their random assignment to their corresponding task (i.e., evaluating one of the five candidate attributes as **Table 1** below describes) should make the assignment mechanism completely orthogonal (i.e., uncorrelated) to the characteristics of the rater himself/herself and the characteristics of the candidates that we asked the rater to evaluate. In the edited version of the paper, we explain these procedures in detail (see relevant paragraphs below).

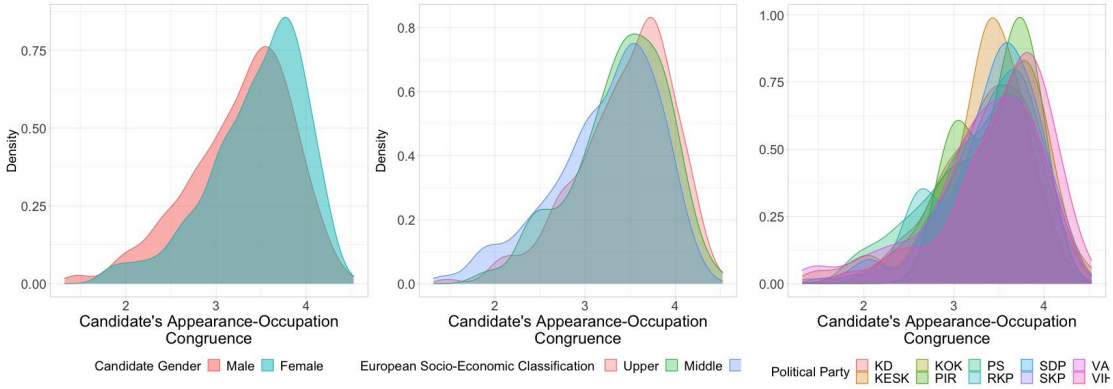
2. **Table 1: Attributes Assigned to Raters by Sampling Criteria and Answer Set**

Sample	Attribute	Question	Answer Set
1	Occupation-Congruent Appearance	*To what extent does this person correspond to your image of someone working in [occupation]?*	5 = perfectly corresponds to my image; 4 = corresponds well to my image; 3 = somewhat corresponds to my image; 2 = does not correspond well to my image; 1 = does not correspond to my image at all.
2	Attractiveness	*In your opinion, how attractive does this person look compared to others of the same age and gender?*	5 = very attractive; 4 = more attractive than average; 3 = average; 2 = below average; 1 = well below average (e.g., Bono et al., 2017; Griffin & Langlois, 2006; Tu et al., 2022).
3	Masculinity	*To what extent do you think that this person looks masculine compared to others of the same age and gender?*	From 1 = not masculine at all to 5 = very masculine (e.g., Hoss et al., 2005).
4	Femininity	*To what extent do you think that this person looks feminine compared to others of the same age and gender?*	From 1 = not feminine at all to 5 = very feminine.

Furthermore, to ensure that R1's comments have been fully considered, we now provide intraclass correlation scores for both male and female candidates. We find that our data show high levels of intraclass correlation, similar to what others have found (see relevant paragraph below, which includes some references).

Overall, and continuing with R1's question, this is the reason we have a "**strong convergence in voters**' **evaluation [about] their occupation judgment**" (as R1 suggests). As Figure 1 shows on p. 10 of the edited paper, we find our strategy to be adequate since the candidate scores are consistent and centered regarding all of the attributes, including the candidates' occupation attributes (see the detail of the figure below). In particular, we find highly consistent levels regardless of gender (**Panel A**), social class (**Panel B**), and party of the candidate (**Panel C**).

<u>Detail of Figure 1 showing Panels A, B, and C (the complete Figure 1 is on p. 10 of the edited version of the paper)</u> :



<u>Edited paragraphs that explain in detail our measurement procedure (pp. 8–9 in the edited version of the paper)</u>:

"In addition, we constructed a novel dataset of physical-appearance ratings using a representative sample of Finnish citizens. First, we drew a random sample of 26,500 Finnish-speaking Finns, aged 18–64, from the Finnish population census. During the fall of 2020, these potential respondents were sent one postal invitation and one postal reminder to participate in an online survey. Respondents received a different survey URL in the invitations. In total, 7,920 raters participated in the survey.

Ideally, we would have directly asked the 7,920 raters to evaluate whether candidates looked as if they came from the working, middle, or upper class. However, that would have introduced social-desirability biases into our responses. This widely studied effect appears in other contexts where subjects avoid answering truthfully due to social pressure, especially when the questions ask about drug use (Druckman et al., 2015), sexual preferences (LaBrie & Earleywine, 2000), attitudes toward race (Kuklinski et al., 1997; Redlawsk et al., 2010) or vote buying (Blair & Imai, 2012; González-Ocantos et al., 2012; Bahamonde, 2022). We believe that directly asking raters to make inferences about the candidates' social classes based on physical appearances constituted an inappropriate question in the Finnish context. Furthermore, the term "social class" is rarely used among the general population because of the "equal Finland" narrative. Thus, to elicit truthful answers, our empirical strategy considered instead asking this question indirectly, that is, asking raters to evaluate the extent to which a member of a category (in this case, a person representing a certain occupation) fits a more generally shared idea or image

of the meaning of the category name (cf. Rosch & Mervis, 1975), only then matching those occupations within the three social classes.

To avoid fatigue effects, we asked raters to assess a random subsample of candidates according to several attributes (see Table 1). Sample 1 evaluated the congruence between the candidate's physical appearance and occupation, Sample 2 evaluated the candidate's attractiveness, Sample 3 evaluated the candidate's masculinity,[4] and Sample 4 evaluated the candidate's femininity.[5] Consequently, the attribute on which raters evaluated their random sample of photographs differed among the four samples. In practice, each of the 7,920 raters evaluated 50 candidate photographs assigned at random.

Importantly, data from the raters show high levels of intraclass correlation: 0.24 for male and 0.22 for female candidates for occupation-congruent appearance, 0.14 for male and 0.22 for female candidates for femininity, 0.15 for male and 0.10 for female candidates for masculinity, and 0.28 for male and 0.35 for female candidates for attractiveness (anonymized). All in all, correlations are consistent with those in widely used data, such as the Add Health data, where the attractiveness correlations range between 0.17 and 0.20 (McClintock, 2014). "

We truly hope that the added paragraph and these detailed responses meet the expectations of R1. However, if you think we should clarify things even more, we would happily do so. Thanks again for your constructive criticism.

4. **What is the connection between perceived occupational status and real occupational status? In other words, do working class candidates also look more working class?**

   Thank you. We think this is an excellent request for clarification, and we see it as an opportunity to provide greater internal validity to our claims. Ideally, we would like to see two things. First, (1) Finnish voters perceive working-class candidates as working-class individuals, middle-class candidates as middle-class individuals, and so on. Otherwise, it would mean that voters have incorrect perceptions of Finnish candidates. If that were the case, it would seriously compromise our empirical claims. Second, (2) we would like to see that candidates who look like they have working-class occupations are actually working-class individuals, and so on. Similarly, having it otherwise would mean that there is a serious mismatch between class and appearance, again compromising our empirical claims.

   To make sure that (1) and (2) hold, we show in a newly added figure that Finnish candidates are scored with an average of 3.3 (a score of 5 is the maximum congruence between physical appearance and occupation). In simple terms, Finnish

[4] Masculinity was defined for respondents using the definition of the official *Dictionary of Contemporary Finnish* (*Kielitoimiston sanakirja*), a dictionary of standard Finnish compiled by the Institute for the Languages of Finland: "masculinity refers to manly, mannish." The respondents were also reminded that both men and women can be perceived as masculine.

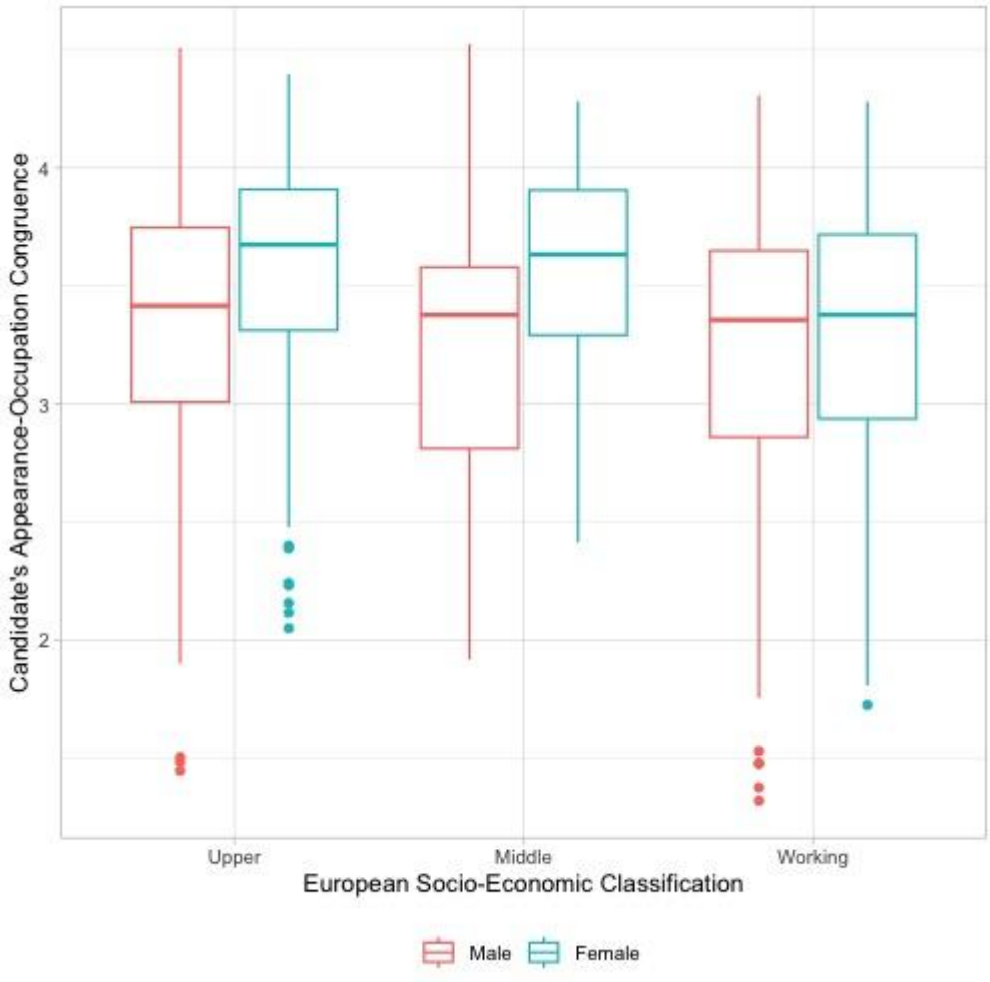
[5] Femininity was defined as "womanly, effeminate" according to the official *Dictionary of Contemporary Finnish*. The respondents were also reminded that both men and women can be perceived as feminine/masculine.

voters find that 66% of the time, candidates do look like they belong to the class to which they actually belong. If we take the r2 statistic as a reference, this would be similar to an r2 of .66, a score traditionally considered “high.” The figure also shows that candidates who look like they have working-class occupations are actually working-class individuals, and so on. Importantly, these levels are consistent for both male and female candidates.

Thus, to answer R1’s question more directly (i.e., “**do working class candidates also look more working class”**), the answer is “yes”: working-class candidates (X-axis) look more working-class (Y-axis), and the same applies to the other two classes.

We sincerely hope that this new figure, the additional paragraph in the revised version of the paper (see below in blue), and the explanatory figure note answers R1’s question. If you think that additional analyses are required to make a stronger case for consistency, we would gladly provide them. We thank you once again for the comment.

<u>Added Figure 2 and an additional explanation in the edited version of the paper (p. 11)</u>:



Added paragraph in the edited version of the paper (p. 11):

"For internal consistency, we would also like to see Finnish voters perceiving working-class candidates as working-class individuals, middle-class candidates as middle-class individuals, and so on. Similarly, we would like to see that Finnish candidates who look as though they have working-class occupations actually are working-class individuals, candidates who look as though they have middle-class occupations actually are middle-class individuals, and so on.[6] Other results would mean that either Finnish voters have a hard time assessing the occupation (class) of the candidate based on the candidate's appearance, or candidates systematically look different from what they are. Either way, one or both assumptions failing to hold would seriously compromise our empirical claims..

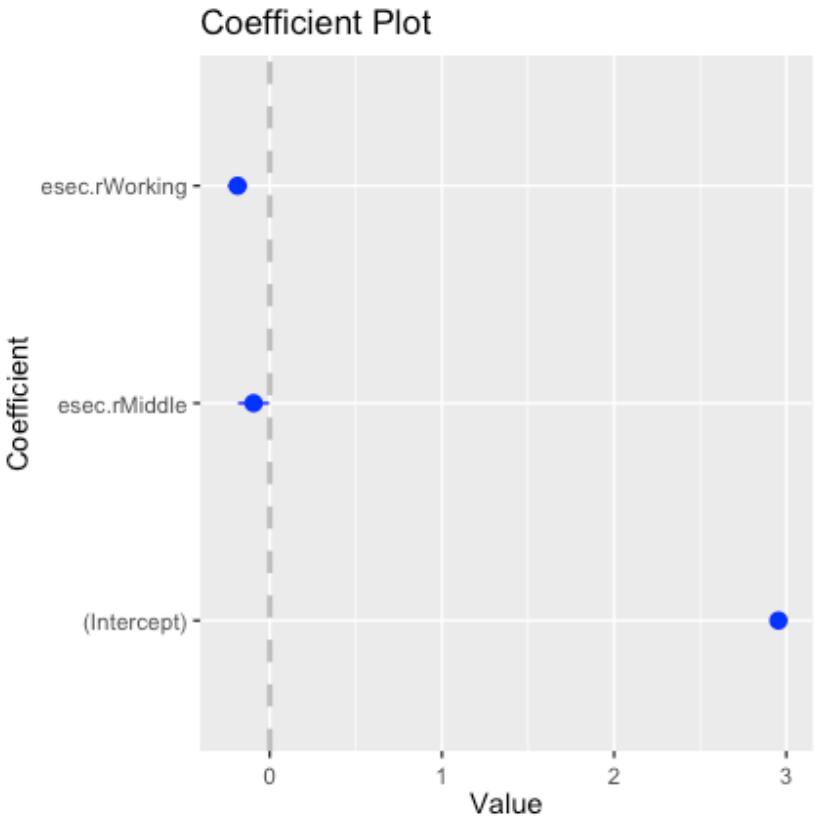
To verify these assumptions, we show in Figure 2 that Finnish candidates scored an average of 3.3 (see Table 1; a score of 5 is the maximum congruence between physical appearance and occupation). This implies that Finnish voters find that 66% of the time, candidates do look like they belong to the class to which they actually belong. The figure also shows that candidates who look like they have working-class occupations are actually working-class individuals, and so on. Importantly, these levels are consistent for both male and female candidates."

6. **I wonder what the correlation between attractiveness and occupational status is?**

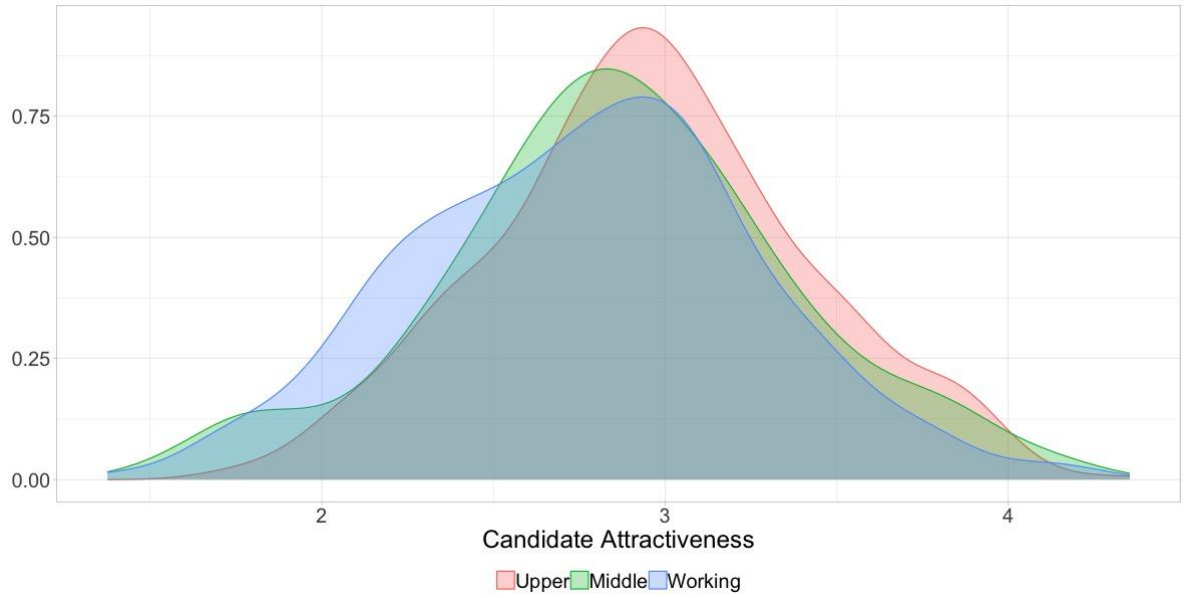
Thank you for this question. In the motivation for the paper (p. 2), we state that the literature has convincingly shown that attractiveness has played an important role in explaining voting intention. That is why we have made sure to control for attractiveness in the original (and edited) version of the manuscript (**see models 4, 6, 7, 8, 11, and 12 in Table 3, p. 17)**. Thus, we find the question relevant to our paper.

To answer the reviewer's question more precisely, **the correlation between attractiveness and occupational status is zero**. The figure below shows a simple linear OLS model that correlates the attractiveness levels of the candidates as a function of their occupational status. In particular, the figure shows beta coefficients that represent the correlation between these two variables (with "upper class" as the reference category).

[6] We thank one of the reviewers for encouraging this discussion.



We do not include this bivariate correlation plot in the paper (but if the reviewer wants us to do so, we would happily do it). What we already had in the paper was **Panel E of Figure 1 (p. 10)**. (See figure detail below for quick reference.) The figure conveys the same information: There is no association between class and attractiveness.



We believe that these null results stress two important ideas. First, there are no self-selection biases in our data-generating process. Otherwise, it would mean, for example, that high-status occupations are systematically perceived as more attractive. In other words—and continuing with the same example—attractive candidates might self-select themselves into high-status occupations ("upper-class"

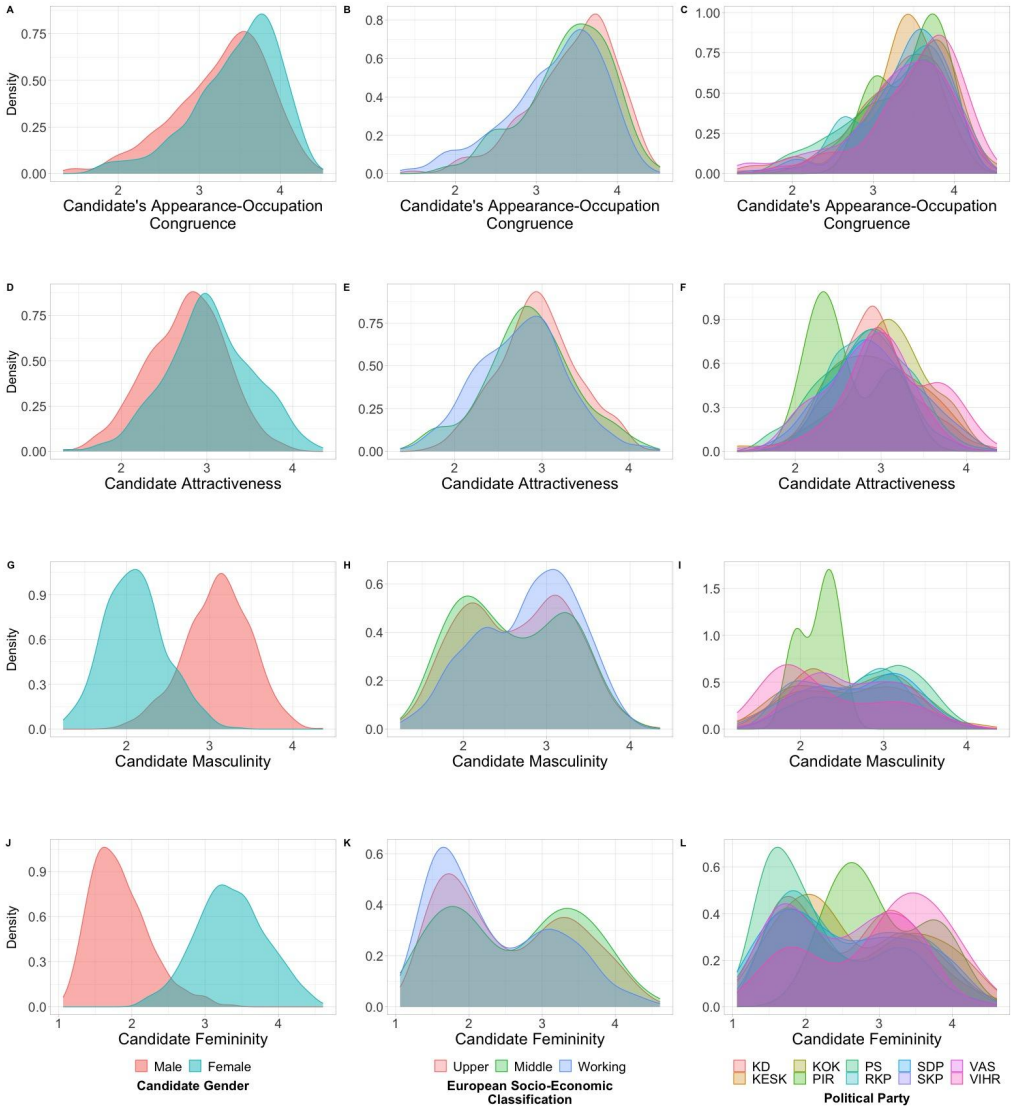
occupations, in the figure above). Second, since we are including both covariates in the same equation (e.g., attractiveness and occupational status), we would like both processes to be orthogonal, i.e., mutually statistically independent). Having it otherwise would violate the statistical assumption of statistical independence, seriously inflating our standard errors (i.e., multicollinearity issues).

To be sure, we now discuss this issue more efficiently in the paper (p. 10 in the paper). See also the figure (same page) we had included in the original submission of our manuscript for reference (which we paste here for quick reference):

"This strategy might raise a concern about self-selection bias if, e.g., high-status-occupation candidates self-select themselves into high-status occupations; or if, for example, "politicians on the right look more beautiful" (Berggren et al., 2010, 2017; see also, Price et al., 2011; Belmi & Neale, 2014; Olivola et al., 2018).[7] Figure 1 presents several distributions after averaging all raters' scores of the 1,415 candidates (see also Rosar et al., 2008, p. 71; Berggren et al., 2010, p. 11). The figure shows these densities broken down by relevant criteria. Since none of the subcategories differ statistically between these groups, the figure strongly suggests that self-selection biases are not a concern in our study."

[7] We thank one of the reviewers for encouraging this discussion.

<u>For reference, we have included Figure 1 below (already included in our original submission). Please see Panel E, also referenced above)</u>:



We do hope the edited paragraph (in blue above) and Figure 1 answer the reviewers' comment. As we now note in footnote #13, we acknowledged the reviewer's comment for encouraging this discussion in the edited version of the paper. Thank you once again for your comment.

7. **The models are rather hypothetical, as they do not control for many important features such as electoral capital, party ideology, campaign spending, whether the seat is safe or not etc... I wonder how strongly the perceived occupational status matters after we control for these features.**

Thank you for this suggestion, and we apologize in advance for the confusion. Our models have already considered those factors. We state in the note with regression tables 2 and 3 (pp. 16-17) that we considered the party of the candidate plus city fixed effects in all models. While the models do consider both sets of variables, we excluded them from the table (for the purpose of streamlining the table's

appearance).

First, controlling for party should suffice to capture the statistical effect of "**party ideology**" (the variable R1 suggested). While a really good suggestion, we consider it best to consider party fixed effects because they enter into the models as separate intercepts without making any empirical assumptions, for example, about where those parties locate in the ideological spectrum or whether they can be classified at all in the ideological spectrum (for example, the "Finns Party" is considered a populist party, which raises concerns if one's purpose is to classify them in the traditional left-right spectrum). To check the actual parties in our sample, please refer to footnote #18 on p. 13 of the manuscript.

Second, to capture the statistical effects of "**electoral capital [...] campaign spending, [and] whether the seat is safe or not etc."** (as suggested by R1), we had adopted the standard empirical strategy of including city fixed effects. We think that R1 is completely right in that we should control for those effects. At the same time, we believe that such factors as electoral capital, campaign spending, and whether the seat is safe or not, among others, are particular to each city. In effect, each city will likely have its own electoral dynamics, with its own contestability and campaign structures, history, and strength of incumbents, and in the case of Finland, language and city/urban cleavages, among many other sources of heterogeneity.

We believe the advantage of having city fixed effects in all our models is that it controls not only for the factors R1 correctly mentions but also for other hard-to-measure but potentially important sources of heterogeneity. The literature has widely used this strategy before. For example, Hainmueller and Hangartner, in "Who Gets a Swiss Passport?" (*American Political Science Review*, 2013, 107(1), 159–187), explain that they use city "**fixed effects so that the effects of the applicant characteristics are identified purely based on the within-municipality variation**" (p. 12). We follow them in that our vector of city fixed effects captures all political, cultural, economic, and geographical effects particular to every city (see Angrist and Pischke, "Mostly Harmless Econometrics," for a reference).

<u>What follows is a paragraph that was (and still is) in the original version of the paper on p. 13</u>:

" Finally, to account for unobserved but fixed omitted variables, we included city fixed effects in all models. By pursuing this econometric strategy, the model aims at controlling for city-specific factors, such as population size, different political cleavages (e.g., urban/rural, language) local party competition issues, and possible incumbency/contestation dynamics, among other hard-to-measure confounding factors (Angrist & Pischke, 2009). "

We truly hope that R1 is satisfied with this set of answers. If not, we are truly open to discussing other possible alternatives. Once again, we thank you for your important question.

8. **The authors should discuss whether they use real or hypothetical candidates. If they use real candidates, face recognition could skew peoples' evaluations**.

We believe this is another excellent question. In the original version of the paper (and the revised version as well), we go *in extenso* about discussing our data. We pasted the relevant sections below (making sure to reference the page so that R1 can cross-check our responses). Thus, we limit this paragraph to give short answers.

Yes, we used real candidates (see longer explanations below in blue). Also, face recognition is a real threat, as R1 correctly points out. To deal with these potential sources of bias, we designed proprietary software that randomly assigned candidate faces to Finnish voters. The software dealt with face recognition because (1) it "**contrasted the candidate's electoral district with the respondent's electoral district and ruled out pictures of candidates from the respondent's electoral district**," and also (2) "**to further minimize recognition, images for which over half of the respondents chose the option 'I recognize the person in the photograph' were excluded from the analysis (N = 4). These included political "superstars," which respondents seemed to recognise based on their past membership in the Finnish Parliament**." See detailed explanation below.

<u>We used real candidates (see pp. 7–8 in the paper)</u>:

"We exploited several data sources. First, in the context of Finnish municipal elections, political campaigns use headshots of the candidates running for office, along with the candidate's name, party affiliation, and occupation. This information is public and appears in city centers, bus stops, public squares, and other highly frequented places, including voting booths (Berggren et al., 2010, p. 12). Importantly, newspaper ads also display the *same* photos (Berggren et al., 2010, p. 10), and "each municipality is obliged by law to provide each party with the same number of slots for posters" (Berggren et al., 2017, p. 83). Second, the Department for Democracy and Public Law (http://www.vaalit.fi) collects this visual information, along with each candidate's vote count, electoral district, age, and occupation (declared by the candidates). Thus, for the 2017 Finnish municipal elections, a subsample of candidates was collected. Out of a universe of 33,618 candidates, we collected over 10,000 photographs.[8] From among the available pictures, we randomly selected 1,500 images, a subsample to maximize statistical representativeness at the national level regarding gender and occupation and, particularly, the share of women in any given occupation. To satisfy the latter criteria, we analyzed data from the Official Statistics of Finland Office (2017) in parallel. The final dataset consisted of 1,415 photographs. Then, taking the candidates' occupational data as a starting point, we followed the European Socioeconomic Classification (Harrison & Rose, 2006) and

[8] Although in principle all photographs ought to be properly archived and publicly available (Act on Cultural Resources in Finland [1433] 2007), gathering the photographs proved laborious. They were processed in GIMP to remove any identifying information. Additionally, to test the usability of the images, we conducted a pilot study, removing images of insufficient quality from the sample. However, Lenz and Lawson (2011) explain that the physical appearance effect holds even when "differences in image quality and other aspects of the pictures, such as visible light, are taken into account" (p. 575).

coded each candidate's occupation by class (i.e., "upper," "middle," or "working class")."

How we dealt with face recognition (see pp. 11–12 in the paper):

"Moving forward to randomly assign pictures to raters, a survey rating tool was specially developed to ensure anonymization. The software presented respondents with random photographs from the pool of images, favoring images that had been rated fewer times. The software contrasted the candidate's electoral district with the respondent's electoral district and ruled out pictures of candidates from the respondent's electoral district. In addition, to further minimize recognition bias, images for which over half of the respondents chose the option'I recognize the person in the photograph' were excluded from the analysis (N = 4).[9] Also, to minimize spillover effects (i.e., to isolate evaluations between the different photographs), a neutral screen with a simple black university logo on a white background appeared. Our design set no time limit for evaluating each picture.[10]"

We truly hope these paragraphs (already included in the original version of the paper) satisfy R1's comment. If R1 would like us to expound on these two issues further, we would happily do so. Thanks again for your suggestions.

Overall comments: Thank you very much for your comments and suggestions. We have answered and implemented every single one of them. We hope they are satisfactory to you. Please let us know if there is anything else we can do.

Reviewer #2

We wanted to thank you for your comments and suggestions. We believe your constructive criticism greatly improved our paper. Below you will find that we have done all we could to implement **all** your suggestions.

1. As you can see, we have grouped the streamlining suggestions in one point. "**I would suggest streamlining the introduction and theory to the main contribution." "[R]elegate data processing choices to the methods section." "I**

[9] These included political "superstars" whom respondents seemed to recognize because of their membership in the Finnish Parliament.

[10] The literature on candidate evaluation consistently affirms that "photographs allow voters to form first impressions" about candidates (Banducci et al., 2008, p. 904). Importantly, voters do not update their initial inferences, even when new information is available (Antonakis & Dalgas, 2009, p. 1183). This "perceptual bias" has been identified in studies with 1-second exposure to facial appearance (Todorov et al., 2005, p. 1623). In fact, Mattes et al. (2010, p. 43) find that "judgments made at 100 milliseconds correlated highly with judgments with no time constraints." Antonakis and Dalgas (2009) report that children have the same (and sometimes even *better*) predicting skills when choosing winning political candidates. Therefore, perceptual biases are invariant to the duration of exposure and are not corrected by increasing levels of political knowledge or age.

would suggest starting with [the role of "physical appearance as social class"] section and then only include what is relevant from the existing literature on heuristics and biases to highlight this contribution." "Outlining separate sections with only the most relevant content would be great" (i.e., do not "mix presentation of the case, the data, analyses and results," "the Finnish case" which "includes both case selection, data sources, and discussion of operationalization of constructs" nor "The section "statistical analysis" [which] includes both results and discussion of methodology").

We think this is an excellent suggestion and have implemented it **<u>in full</u>**. As this involved streamlining the whole paper, for simplicity, we do not reproduce the edited paragraphs here in the memo. If necessary, the reviewer is kindly invited to check the edited version of the paper to cross-check our response.

2. "**It would probably be easier for a reader to follow the models if the baseline models are models without any interactions and then add these in a second step**."

Yes–thanks so much for this suggestion. We think it adds more transparency to our results. **We have added a new table** (Table 2 on p. 16) with the same models shown in the original table (now Table 3 in the edited version of the paper, see p. 17) but without the interaction term. Please see Table 2 below in this memo.

As Table 2 shows, **our main variables of interest remain statistically significant on their own, regardless of the controls**. They also show the expected signs. Notably, all these specifications (similar to the models in the main table) include party and city fixed effects.

<u>New table added. Table 2 now shows what R2 suggested, i.e., all specifications without the interaction term</u> (p. 16 of the revised version of the paper):

	1	2	3	4	5	6	7	8	9	10	11	12
	Full	Male	Female	Full	Full	Full	Male	Male	Male	Female	Female	Female
Intercept	5.08 ***	3.40 ***	6.63 ***	2.41 ***	4.90 ***	4.93 ***	0.16	3.34 ***	1.68 ***	3.51 ***	6.89 ***	3.27 ***
	(0.05)	(0.08)	(0.06)	(0.05)	(0.05)	(0.05)	(0.08)	(0.08)	(0.08)	(0.07)	(0.06)	(0.08)
Appearance-Occupation Congruence	0.26 ***	0.40 ***	0.16 ***	0.15 ***	0.27 ***	0.25 ***	0.25 ***	0.40 ***	0.40 ***	0.13 ***	0.15 ***	0.18 ***
	(0.00)	(0.01)	(0.01)	(0.00)	(0.00)	(0.00)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Middle Class	-0.33 ***	-0.35 ***	-0.47 ***	-0.31 ***	-0.33 ***	-0.33 ***	-0.22 ***	-0.35 ***	-0.28 ***	-0.50 ***	-0.47 ***	-0.53 ***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Working Class	-0.53 ***	-0.44 ***	-0.78 ***	-0.44 ***	-0.54 ***	-0.52 ***	-0.33 ***	-0.45 ***	-0.36 ***	-0.74 ***	-0.77 ***	-0.70 ***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Age	-0.02 ***	-0.01 ***	-0.03 ***	-0.00 ***	-0.02 ***	-0.02 ***	0.00 ***	-0.01 ***	0.00 ***	-0.01 ***	-0.03 ***	-0.02 ***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Attractiveness				0.78 ***			1.03 ***			0.73 ***		
				(0.01)			(0.01)			(0.01)		
Masculinity					0.08 ***			0.02 **			-0.19 ***	
					(0.00)			(0.01)			(0.01)	
Femininity						0.05 ***			0.54 ***			0.69 ***
						(0.00)			(0.01)			(0.01)
AIC	289962.26	172895.19	79620.46	268840.99	289492.30	289673.94	155861.99	172888.00	168386.44	73501.59	79392.16	74575.78
BIC	291184.00	173850.85	80436.78	270067.78	290719.26	290900.90	156822.34	173848.35	169346.78	74321.92	80212.82	75396.44
Log Likelihood	-144747.13	-86243.60	-39622.23	-134185.50	-144511.15	-144601.97	-77726.00	-86239.00	-83988.22	-36561.80	-39507.08	-37098.89
Deviance	280978.73	167552.51	75663.61	259861.79	280506.76	280688.40	150517.31	167543.32	163041.75	69549.08	75433.31	70616.93
Num. obs.	1368	800	568	1367	1368	1368	800	800	800	567	568	568

*** p < 0.001; ** p < 0.01; * p < 0.05. Dependent variable is the number of votes each candidate received. City fixed effects and party variables omitted. The reference category in the ESEC variable is 'Upper Class.' Given the symmetry of the derivatives, changing the reference category does not alter the interpretation of the results. Functional form is Poisson regression for all models.

We have **also added the paragraph below** (pp. 13–14 in the paper), where we discuss the importance of having such a table—one with all the models but without the interaction term. We made sure to thank you for your contribution in footnote #20 in the paper.

<u>We also added a brief discussion about the new table</u> (pp. 13–14 of the revised version of the paper**<u>)</u>**:

"Before introducing our main results (Table 3 and Figure 3), we show our baseline models in Table 2. These models show the same variables considered in the main regression table but without the interaction term.[11] The idea is to suggest that our main interactive results are not a statistical artifact. As Table 2 suggests, all relevant variables are statistically significant and have a sizable effect and the expected sign."

We thank you for your suggestion and truly hope we answered your comment satisfactorily.

3. "**In the regression table the $times$ also looks weird**."

Yes, we have fixed this issue. Thank you for your suggestion. Please check the table below.

We have edited the main regression table (now Table 3), which no longer has that typo (p. 17 of the revised paper):

	1	2	3	4	5	6	7	8	9	10	11	12
	Full	Male	Female	Full	Full	Full	Male	Female	Male	Female	Male	Female
Intercept	4.49 ***	2.58 ***	6.17 ***	2.11 ***	4.27 ***	2.15 ***	-0.25 **	3.24 ***	2.50 ***	2.98 ***	0.17	2.80 ***
	(0.05)	(0.08)	(0.07)	(0.05)	(0.05)	(0.05)	(0.09)	(0.08)	(0.08)	(0.08)	(0.09)	(0.08)
Appearance-Occupation Congruence	0.41 ***	0.59 ***	0.28 ***	0.25 ***	0.42 ***	0.26 ***	0.37 ***	0.22 ***	0.59 ***	0.28 ***	0.37 ***	0.24 ***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Middle Class	0.30 ***	0.81 ***	-0.05	0.41 ***	0.23 ***	0.31 ***	0.54 ***	0.12	0.77 ***	0.18	0.75 ***	0.16
	(0.06)	(0.08)	(0.11)	(0.06)	(0.06)	(0.06)	(0.08)	(0.11)	(0.08)	(0.11)	(0.08)	(0.11)
Working Class	1.03 ***	1.59 ***	0.70 ***	0.49 ***	1.07 ***	0.58 ***	0.95 ***	0.23 ***	1.59 ***	0.32 ***	0.86 ***	0.19 **
	(0.03)	(0.04)	(0.07)	(0.03)	(0.03)	(0.03)	(0.04)	(0.07)	(0.04)	(0.07)	(0.04)	(0.07)
Age	-0.02 ***	-0.01 ***	-0.03 ***	-0.00 ***	-0.02 ***	-0.01 ***	0.00 ***	-0.01 ***	-0.01 ***	-0.02 ***	0.01 ***	-0.01 ***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Appearance-Occupation Congruence X Middle Class	-0.18 ***	-0.34 ***	-0.11 ***	-0.20 ***	-0.16 ***	-0.17 ***	-0.22 ***	-0.17 ***	-0.33 ***	-0.19 ***	-0.28 ***	-0.18 ***
	(0.02)	(0.02)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.02)	(0.03)	(0.02)	(0.03)
Appearance-Occupation Congruence X Working Class	-0.46 ***	-0.61 ***	-0.43 ***	-0.28 ***	-0.48 ***	-0.31 ***	-0.39 ***	-0.28 ***	-0.62 ***	-0.29 ***	-0.35 ***	-0.27 ***
	(0.01)	(0.01)	(0.02)	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)	(0.01)	(0.02)	(0.01)	(0.02)
Attractiveness				0.76 ***		0.89 ***	1.00 ***	0.72 ***			1.08 ***	0.53 ***
				(0.01)		(0.01)	(0.01)	(0.01)			(0.01)	(0.01)
Masculinity					0.09 ***				0.04 ***		-0.28 ***	
					(0.00)				(0.01)		(0.01)	
Femininity						-0.15 ***				0.68 ***		0.26 ***
						(0.00)				(0.01)		(0.02)
AIC	287653.94	170596.79	79111.45	267973.43	287029.77	265328.79	154959.12	73290.19	170577.24	74331.79	153861.01	73001.39
BIC	288886.12	171561.82	79936.45	269210.66	288267.17	266571.24	155928.83	74119.20	171546.96	75161.13	154835.41	73834.74
Log Likelihood	-143590.97	-85092.40	-39365.73	-133749.72	-143277.88	-132426.40	-77272.56	-36454.09	-85081.62	-36974.89	-76722.51	-36308.70
Deviance	278666.41	165250.11	75150.60	258990.24	278040.23	256343.60	149610.43	69333.68	165228.56	70368.94	148510.33	69042.88
Num. obs.	1368	800	568	1367	1368	1367	800	567	800	568	800	567

*** p < 0.001; ** p < 0.01; * p < 0.05. Dependent variable is the number of votes each candidate received. City fixed effects and party variables omitted. The reference category in the ESEC variable is 'Upper Class.' Given the symmetry of the derivatives, changing the reference category does not alter the interpretation of the results. Functional form is Poisson regression for all models.

4. "**[S]ince [the equation is] not included it is in fact difficult to exactly see what is being estimated and what is being calculated**."

Thank you, we could not agree more. We have now introduced **Equation 1,** which represents the main model (models 1, 2, and 3 in Table 3; please see p. 13 in the paper).

We have added the main equation that was estimated (see Equation 1 on pp. 12–13) plus a brief description:

"Following the statistical literature, we identify the data-generating process of the dependent variable as a Poisson process (Long, 1997).[12] […] More formally, we fit the following Poisson model

[11] We thank one of the reviewers for this suggestion.

[12] In Poisson processes, the mean of the outcome variable equals the variance of the outcome variable (Long, 1997, p. 218). In our case, voting shows some evidence of overdispersion. To relax the equidispersion assumption, both negative binomial and quasi-Poisson specifications were estimated (Fox & Weisberg, 2011). The substantive results remain intact. These analyses are available upon request.

$$\log(\text{Votes}_i) = \beta_1 \text{Occupation-Appearance Congruence}_i \times \text{Social Class}_i + \beta_2 \text{Age}_i + \gamma 1 \text{Party}_i + \gamma 2 \text{City}_i + \Theta_i \quad (1)$$

where the number of Votes of every candidate *i* is estimated as a function of the interaction between the levels of Appearance-Occupation Congruence and Social Class for every candidate *i* (captured by coefficient), Age of candidate *i* (captured by coefficient), Party of candidate *i*, and the City where candidate *i* ran (captured by fixed effects and respectively). In addition, we considered extended control variables (i.e., attractiveness, masculinity and femininity levels) in the vector of observables. We also estimated several other combinations as robustness checks."

We hope these answers clarify your questions. We thank you once again for your constructive criticism.

5. "**[I]t is typical to distinguish between** '**marginal effects,**' **the partial derivative, and** '**predicted values,**' **simply plugging values into the estimated equations using interesting values for all variables. Here we are presented with** '**predicted marginal effects**' **which is almost self-contradictory**."

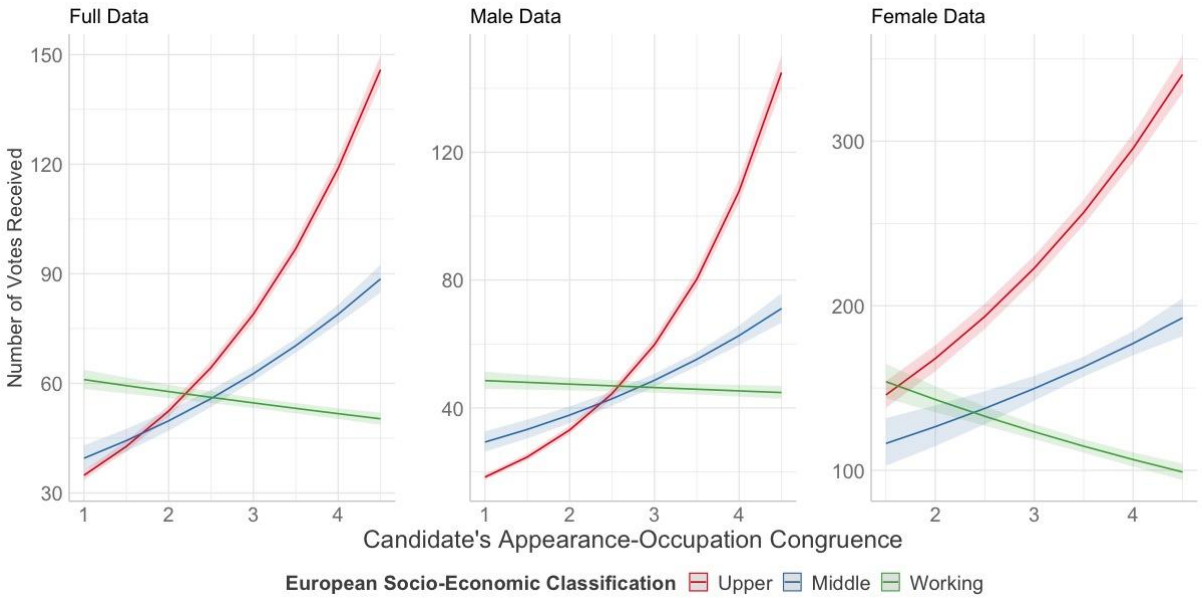
You are absolutely correct, and we apologize for the confusion. **We always meant** "**marginal effects." To clarify the issue even further, we have done two things. First, we reference the R package that computed the marginal effects** (see added paragraph below). The package clearly states that it calculates and plots the marginal effects (if necessary, see online manual of R package in PDF, p. 28 in the online manual). **We think that referencing this package will add more clarity to the calculations our paper presents**. Second, we have modified the title and axis of Figure 3 and made it explicit that we are talking about marginal effects. Thus,

We have referenced the R package so our readers know exactly the mechanics of the calculations being made (see p. 12 of the paper):

"Following Brambor et al. (2006), the equation included both the multiplicative and constitutive terms. We also analyzed the marginal effects plot (in Figure 2; see Lüdecke, 2021)."

We have modified the title and axis titles of Figure 3 (see p. 15 of the paper):

"Figure 3: Marginal Effects of the Interaction Term Between Physical Appearance-Occupation Congruence and Social Class on Votes Received"



We truly hope these responses clarify your point. If not, please let us know, and we will happily provide more details.

Overall comments: Thank you very much for your comments and suggestions. We have answered and implemented all of them and hope we have done so to your satisfaction. Please let us know if there is anything else we can do.